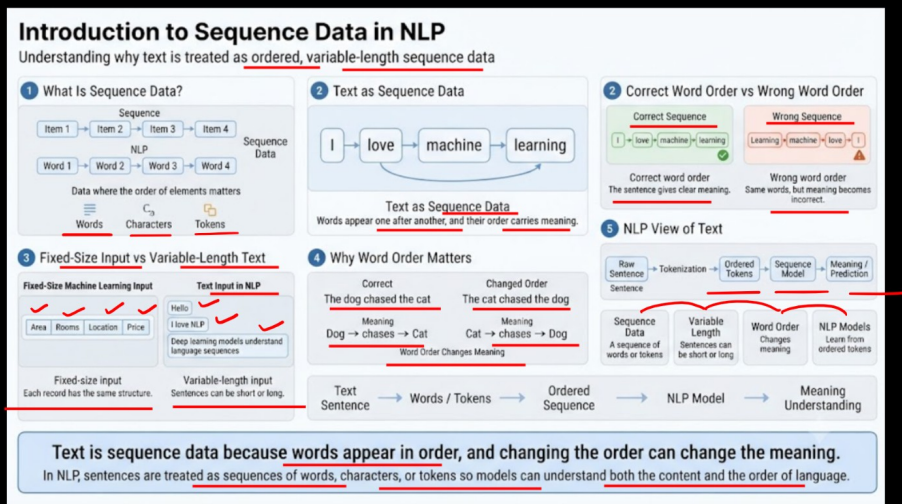
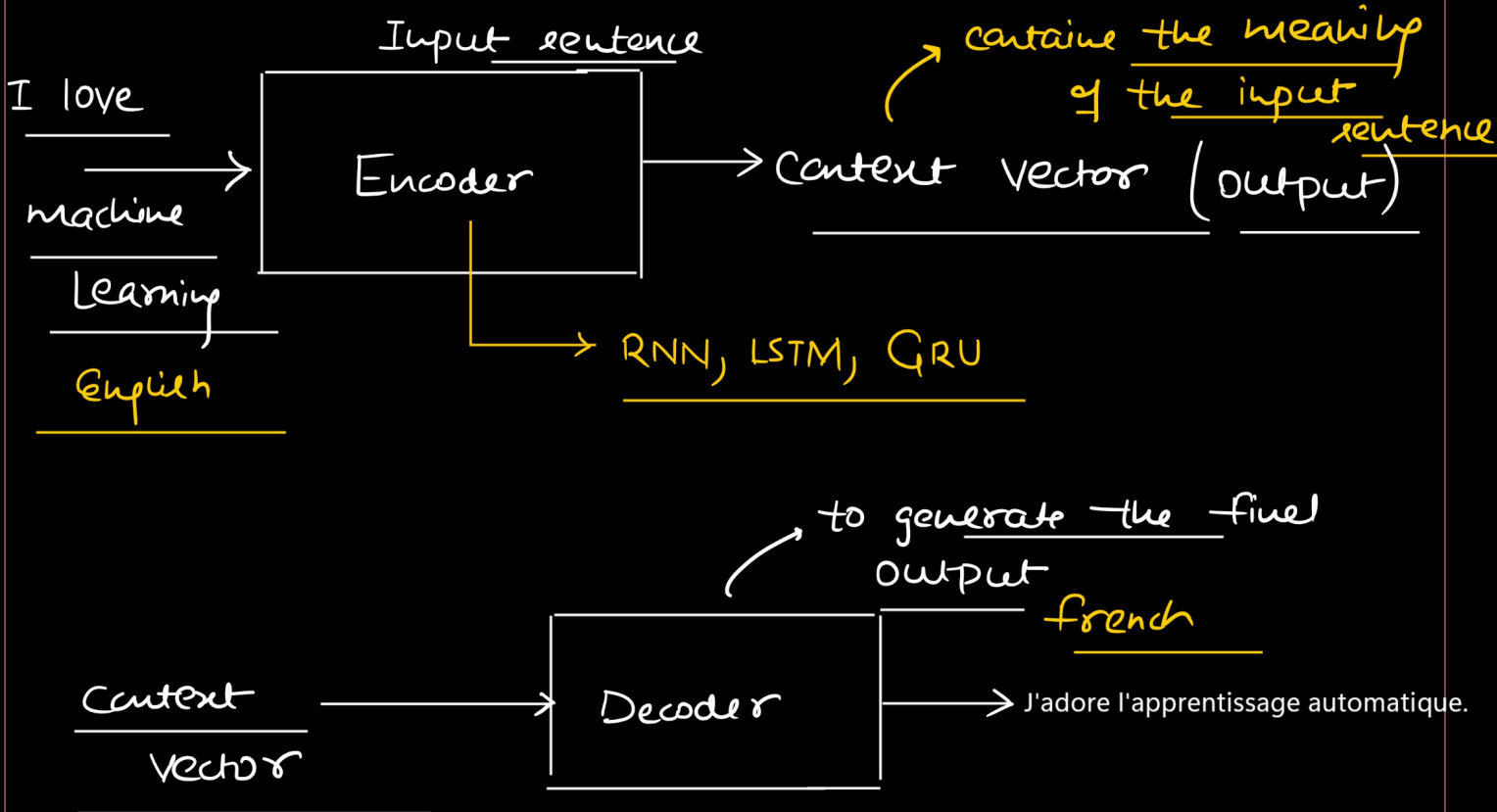


Sequence data



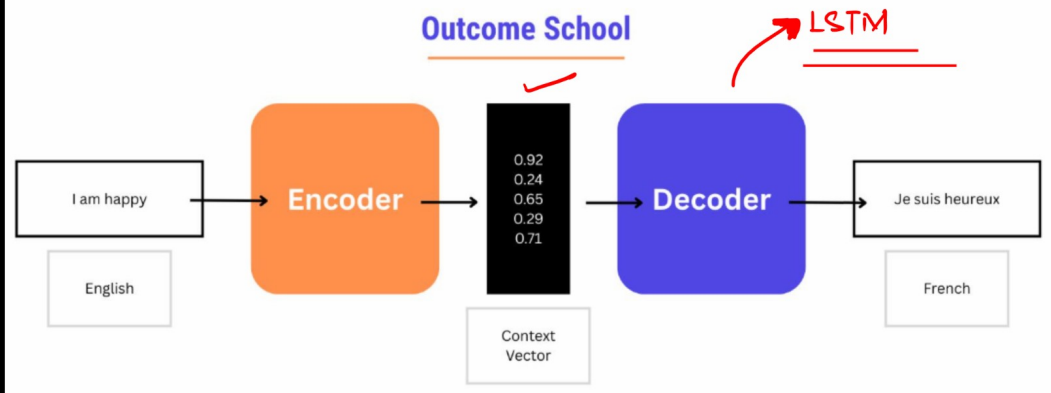
Encoder Decoder Architecture



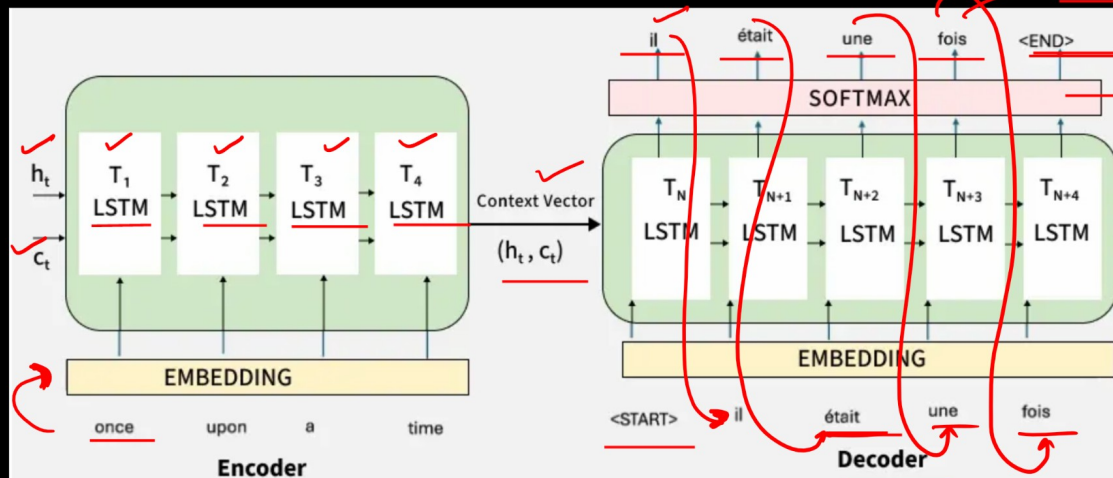
Encoder-Decoder Architecture

Machine Learning

Outcome School

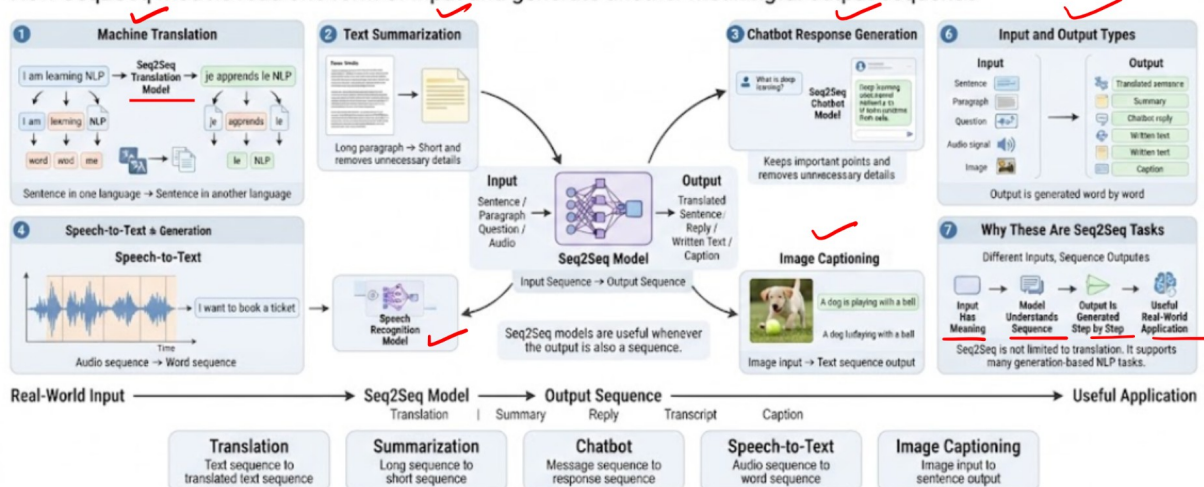


probability
of
next
target
word



Real-World Examples of Seq2Seq Models

How Seq2Seq models read one form of input and generate another meaningful output sequence



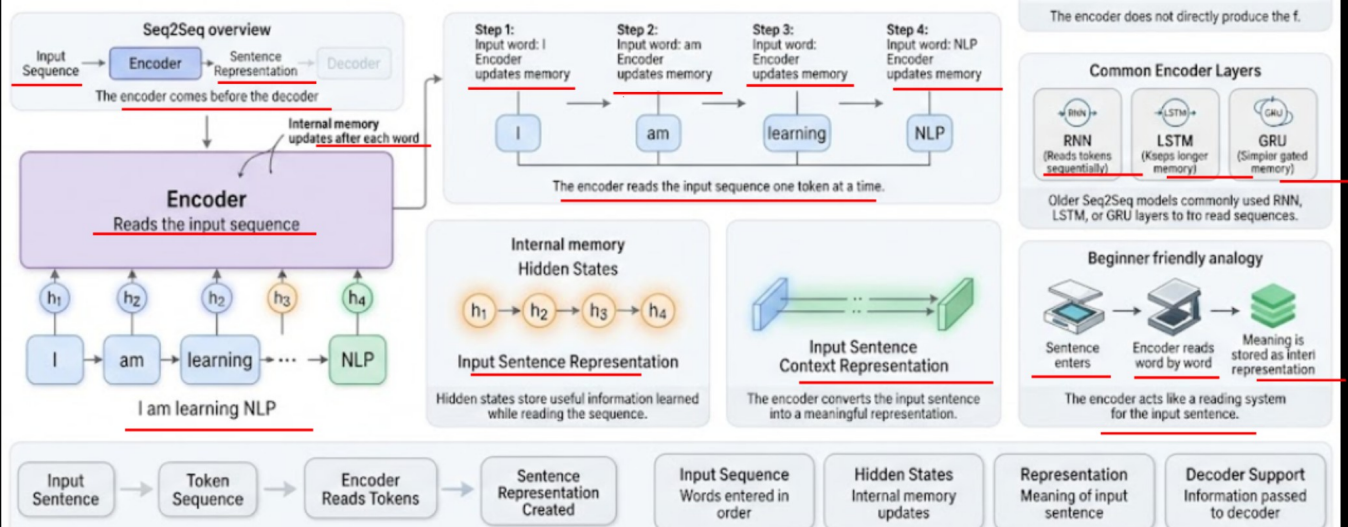
Seq2Seq models are used whenever a system needs to read one input and generate another meaningful output sequence. Machine translation, text summarization, chatbot replies, speech-to-text, and image captioning are common real-world examples where Seq2Seq models convert input information into useful generated text.

Point	Classification Model	Seq2Seq Model
Input	Sequence or fixed data	Sequence / <i>Variable</i>
Output	Single label	Full sequence
Output length	Fixed	Variable
Example task	Sentiment analysis	Translation
Output style	One class	Word-by-word generation

Positive / Negative

What Is an Encoder in a Seq2Seq Model?

How the encoder reads an input sequence and converts it into useful internal information

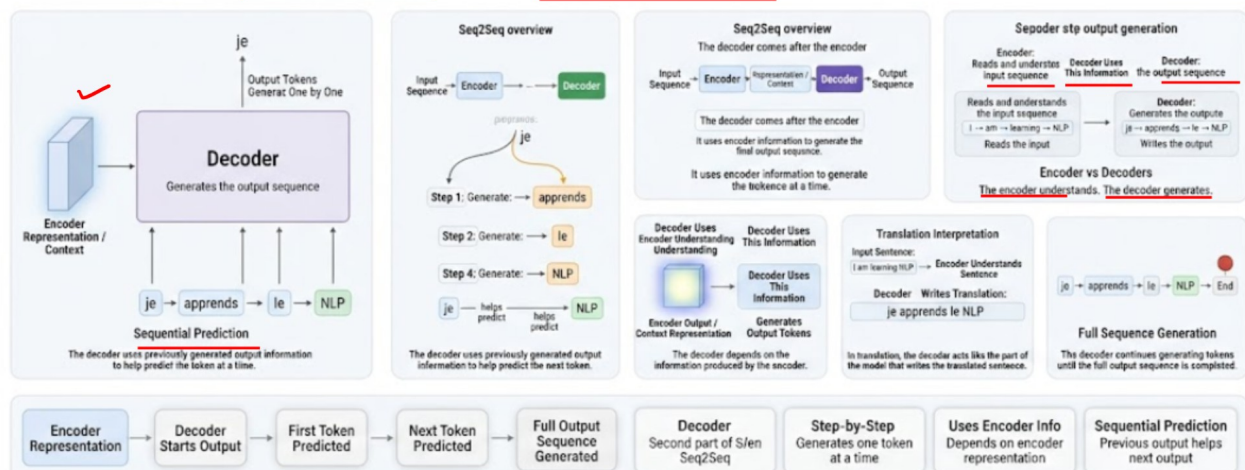


The encoder reads the input sequence word by word and converts it into internal information that the decoder can use.

In a Seq2Seq model, the encoder does not directly generate the final output. Its main job is to understand the input sequence and create a useful representation through hidden states.

What Is a Decoder in a Seq2Seq Model?

How the decoder uses encoder information to generate the output sequence one token at a time



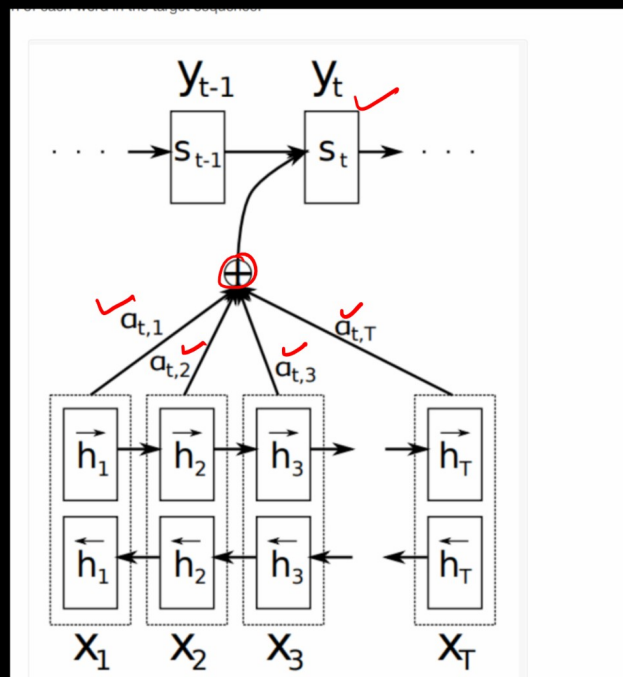
The decoder generates the output sequence one token at a time using the information from the encoder.

In a Seq2Seq model, the decoder is responsible for producing the final output. It predicts one word after another until the complete output sequence is generated.

↳ Decoder does not generate all words
at once.

↳ Predicts the first word, then uses that information to predict the next word & this process continues until the full sequence is generated.

Attention Mechanism



Attention

Encoder 4 $\rightarrow h_4$

Encoder 3 $\rightarrow h_3$

Encoder 2 $\rightarrow h_2$

Encoder 1 $\rightarrow h_1$

Attention

Input $\rightarrow$ I love machine learning.

0.05 0.10 0.60 0.25

machine $\rightarrow$ attention score

$$\text{Context} = \sum_i \alpha_i h_i$$

hidden state

Attention is a mechanism that helps a model focus on the most relevant parts of the input sequence.

Instead of depending only on one final context vector, the decoder can look at all encoder hidden states and decide which input words are more important for the current output word.

